TCG 6223- COMPUTER GRAPHICS

GUIDELINES FOR GROUP PROJECT

Blind Boxes Collection

Description

Create an engaging and visually appealing blind box collection with interactive battle mechanics. You are to build a blind boxes collection using OpenGL consisting of object models and simple animations while applying various rendering effects such as lighting, shading, and texturing.

Each team is required to build TWO blind box characters, which can cast skill in a BATTLE. The characters should be user controlled during battle. Each character should be able to at least cast ONE skill. You may choose any blind box characters and design their unique skill based on their distinctive personality.
For example:
Dimoo: Casts a protective shield spell to block attacks.
Labubu: Sets traps to hinder opponents.

Grouping

Four members per team. No copying is allowed, whether from website or from your other teams.

Deadline

Submit project LATEST by **12 JUNE 2025 11:59pm. Late submission will be deducted 5% each day.**
Upon submission, each team is required to demonstrate their work using own laptop.

Submission of 4 Items

1. Written Report in PDF Format

Each team must prepare a report that introduces and describes the construction process of your object models. Ensure your report is clearly written, neat, and interesting. Nonetheless, it should contain at least the following contents:

a) A Cover Page

- ✓ Contain your Team Name, students' ID, names and Lab Section.

b) Introduction

- ✓ Briefly introduce your object model and its animation to make readers understand your concept. A screen shot of the model should be included for reference.

c) Documentations of Virtual Model

- ✓ Describe the building objects in your virtual model.
- ✓ Explain how the objects are built, and how the animation works, if any.

d) User Manual / Instructions

- ✓ Show how to interact with the program with keyboard and/or mouse, if any.

e) Screenshots

- ✓ A few figures that portray the look of the virtual environment.
- ✓ Provide a short description for each figure.

2. Program

Provide the source code of the program in an organized filename with description documented within the source code. It is your full responsibility to make sure that source code is error-free. Your code must be compatible with **MinGW C++** compiler using **Code::Blocks IDE** under **Windows 10**.

3. Team Declaration Form

Fill up milestones done by each member and show the timeline of project development.

4. Video Clip

Record model demonstration in a LESS THAN 5 minutes video.

Evaluation Criteria

The following are some general guidelines that you should follow to come out with a good and memorable project.

Written Report

a) Try making your report concise, straight to the point, and fun to read. A long report is not equivalent to a good report.
b) Please check for grammatical mistakes!
c) Do not use chatting-like spelling (i.e. dun, gud, hv, gg, =.= etc)!

Program

a) Use consistent indentation, well-documented and organized.
b) Reuse your code as much as possible. NO HARD CODING, please. Use function or class whenever possible.
c) Apply as much knowledge from lab to gain more marks.
d) Optimize your code to ensure program efficiency by using loops or mathematical equations whenever possible.
e) Make sure you are using memory space efficiently as well.

f) The more complex the objects you created on your own (not counting those generated by graphics software, not counting GLUT objects), the more marks you will get!
g) **Build your virtual environment incrementally**. If you want to build an object with moving parts, you should not build the model as a single object to avoid the difficulty of moving the various moving parts later since it has already been fixed to the rest of the model.
h) Each object should be **modeled in its own coordinate system with (0,0,0) being the origin** of the coordinate system. You can **lay out all your objects on the X-Z plane** of the virtual environment first. After that, you can move it to the location using appropriate transformation.
i) **Apply lighting and texture mapping appropriately**. Do not apply things just for the sake of applying.
j) **Simple animation is sufficient** for this project. However, you will be rewarded well if you can come out with nice and complex animations.
k) **Add your own user-interaction** to make your program more interesting.

Bonus

You may be granted bonus marks if you **incorporate any special effect** into your program, especially those which are **not being taught in classes**.

Golden Tips

Here are some tips to help you create a creative and memorable project:

1. **Plan Thoroughly**:
   - ✓ Start with a clear plan outlining your project's goals, timeline, and key milestones.
   - ✓ Break down tasks into manageable steps and assign responsibilities within your group.
2. **Research and Inspiration**:
   - ✓ Research existing blind box characters and their unique traits to inspire your designs.
   - ✓ Look into various rendering techniques and animations to enhance your visual appeal.
3. **Design with Personality**:
   - ✓ Ensure your characters have distinctive personalities and appearances.
   - ✓ Design unique skills for each character that reflect their traits and add excitement to the battle mechanics.
4. **Focus on User Experience**:
   - ✓ Make the characters user-controlled to provide an interactive and engaging experience.
   - ✓ Ensure the controls are intuitive and responsive for a smooth gameplay experience.
5. **Implement Rendering Effects**:
   - ✓ Apply various rendering effects such as lighting, shading, and texturing to bring your characters and environment to life.
   - ✓ Experiment with different techniques to achieve the best visual results.
6. **Test and Iterate**:
   - ✓ Regularly test your project to identify and fix any issues.
   - ✓ Gather feedback from peers to improve your project continuously.
7. **Presentation Matters**:
   - ✓ Prepare a clear and concise presentation to showcase your project.
   - ✓ Highlight the unique aspects of your characters, their skills, and the rendering effects.
8. **Document Your Work**:
   - ✓ Keep detailed documentation of your design process, code, and any challenges you faced.
9. **Stay Creative and Have Fun**:
   - ✓ Enjoy the creative process and don't be afraid to experiment with new ideas.
   - ✓ A memorable project often comes from passion and enthusiasm.

BEWARE!

a) **NO code generator** can be used under any circumstances. There are software tools that can generate OpenGL code from models built using graphics software such as Maya and 3DStudio etc., these tools ARE STRICTLY PROHIBITED.

b) For complicated models, you may use software such as Maya or 3DStudio to generate the models, HOWEVER, **the use of this software is STRICTLY LIMITED to generation of vertices and faces information** (such as coordinates, face list, textures, colors etc) ONLY. In this case, this information must be stored as text files and you must write your own code to read them into your program. You MUST NOT include this information directly into your program.

c) The source of all **downloaded models must be documented and acknowledged**.

d) You are advice that the use of these models will not gain you any advantages or extra marks, however, beautify the models may be. On the contrary, there will be more marks if you model

everything using your own codes. Thus, do not waste your time using this software or downloading ready-made models. It would be fine only if you absolutely need to use them to model something that is just impossible to model using your own codes.

ENJOY YOUR TEAMWORK!

PROJECT MARKSHEET

No.	Description	Max Score (%)
1	**Construction of Characters** The graphics model, vertices list, polygon, object transformation, etc. ** *0-fail 1-3: unsatisfactory 4-7: meet requirement 8-11: above requirement 12-14: excellent*	14
2	**Animation of Characters Skills** Various rendering effects such as lighting, shading, texturing, etc. ** *0-fail 1-3: unsatisfactory 4-7: meet requirement 8-11: above requirement 12-14: excellent*	14
3	**Battle Environment** The environment changes during battle. ** *0-fail 1-2: unsatisfactory 3-4: meet requirement 5-6: above requirement 7-8: excellent*	8
4	**Document Management** Report Formatting Program/ Source code ** *0-fail 1: unsatisfactory 2: meet requirement 3: above requirement 4: excellent*	4
5	**Penalty for late submission (5 marks per day)**	
	TOTAL	**40**